

AI Use, Age, and Mental Health

An exploratory analysis of the Global Mind Project survey

Dataset: gmp_ai_only_2026-04-14.csv — 266,589 respondents who answered the AI-use question (incl. 88,573 non-users as a natural baseline), 205 variables.

Scope: How AI-use *frequency* and AI-use *purpose* relate to mental-health (MHQ) and to other lifestyle, social, and demographic factors — analysed within age groups, starting with young adults (18–24).

Methods: dose-response curves, Wilson/CI group means, Cramér's V association screening, Bayesian-network structure learning (hill-climb + BIC, χ^2 -pruned), and competing-exposure logistic regression.

A note on causation: every result below is *cross-sectional and associational*. Directions are not identified; reverse causation (distressed people using AI differently) is consistent with all findings.

Executive summary

This investigation began with a simple question — does AI-use frequency correlate with the Mental Health Quotient (MHQ) among young people? — and progressively refined it. The headline result is that **what people use AI for matters far more than how often they use it.**

- **Frequency × MHQ is an inverted-U.** At every age, wellbeing rises from non-use to moderate (weekly) use, then falls for the heaviest users. The dip is steep for the young and negligible by age 65+.
- **A Bayesian network isolates the young-specific link.** For 18–24, AI-use sits directly beside anxiety and depression even after conditioning on confounders; for 25+, it connects only to education and employment.
- **It is not just ‘screen time’.** Heavy-AI anxiety risk (OR 1.50) survives adjustment for social-media and gaming. Gaming is, if anything, protective for anxiety.
- **Purpose dominates.** Using AI as a *tool* (productivity, skills, creativity) tracks *lower* anxiety; using it as a *companion* (emotional support, companionship, ‘rely on it’, ‘more isolated’) tracks much higher anxiety — effect sizes 5–10× the frequency effect. Controlling for purpose, frequency’s effect nearly vanishes.
- **The companion link is broad and universal.** It holds at every level of real-world social support and at every age. Young people simply use AI as a companion far more often (39% vs 13% at 65+), which is why the raw effect looked young-specific.

1. Data and method

The source file is the Global Mind Project (GMP) extract restricted to respondents who answered the AI-use question. AI-use frequency was mapped to an ordinal scale (0 = never ... 6 = all of the time); ‘users’ are those at level ≥ 1 . Age was grouped as 18–24, 25–34, 35–44, 45–54, 55–64, and 65+.

MHQ (range –100 to +200; higher = better) and its six sub-scores are computed by GMP. To study symptoms specifically — and because the survey’s PHQ-9/GAD-7/Cantril fields are empty in this extract — we built **depression** and **anxiety** proxies from the 1–9 ‘spectrum’ functioning items, oriented against MHQ (proxy–MHQ correlations –0.76 and –0.78, confirming validity).

AI-use purpose comes from the multi-select fields `ai_use_social` (emotional support, companionship, romantic conversation) and `ai_impact_personal` (self-reported benefits and harms).

2. AI-use frequency vs MHQ: an inverted-U at every age

Among young adults, mean MHQ climbs from non-users to weekly users (+22 points) and then collapses for the heaviest users — the ‘all of the time’ group (MHQ 6.2) is worse off than people who never use AI (24.1). A single linear correlation ($\rho = -0.09$) hides this curve.

AI use (18–24)	n	Mean MHQ
Never	4,284	24.1
Rarely	10,817	36.6
Monthly	6,521	43.0
Weekly (peak)	13,915	45.8
Daily	11,487	36.5
Constantly	10,765	23.5
All the time (worst)	2,070	6.2

Repeating the curve for every age group shows the same shape everywhere, but the heavy-use cliff is deep for the young and flattens with age.

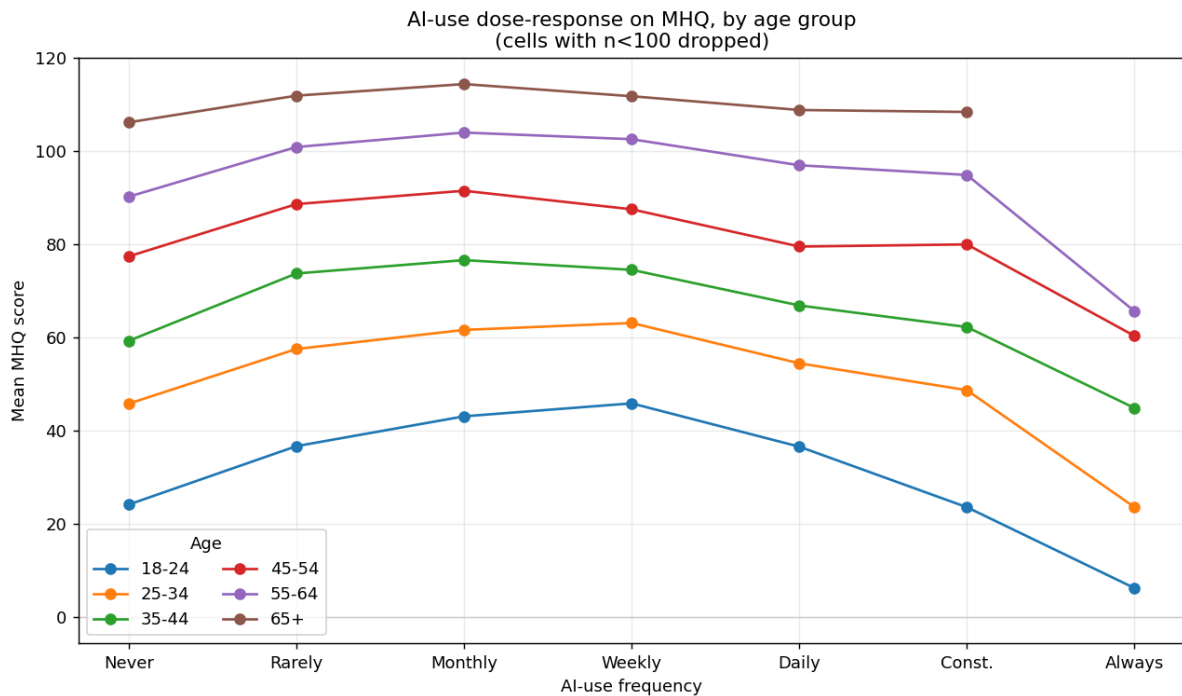


Figure 1. AI-use dose-response on MHQ, by age group. Every cohort peaks at moderate use; the heavy-use decline is largest for 18–24 (blue) and minimal for 65+ (brown).

3. Bayesian network: what AI-use is directly tied to

To separate genuine links from confounding, we learned a Bayesian network over AI-use, the distress outcomes, and ten lifestyle / structural confounders (employment, education, sleep, exercise, diet, social contact, gaming, social-media, substance use, medical condition) on 45,366 complete-case young respondents. The **Markov blanket** of AI-use — its direct neighbours — is the key output.

For 18–24, AI-use is directly tied to anxiety and depression (plus education, employment, social-media). Every other lifestyle factor is conditionally independent of AI-use once education and employment are known. Crucially, the same network learned for older cohorts links AI-use to *only* education and employment — the mental-health coupling is young-specific.

Bayesian network — YOUNG (18-24), n=45,366
red = AI_use orange = distress outcomes red edges = AI_use Markov blanket

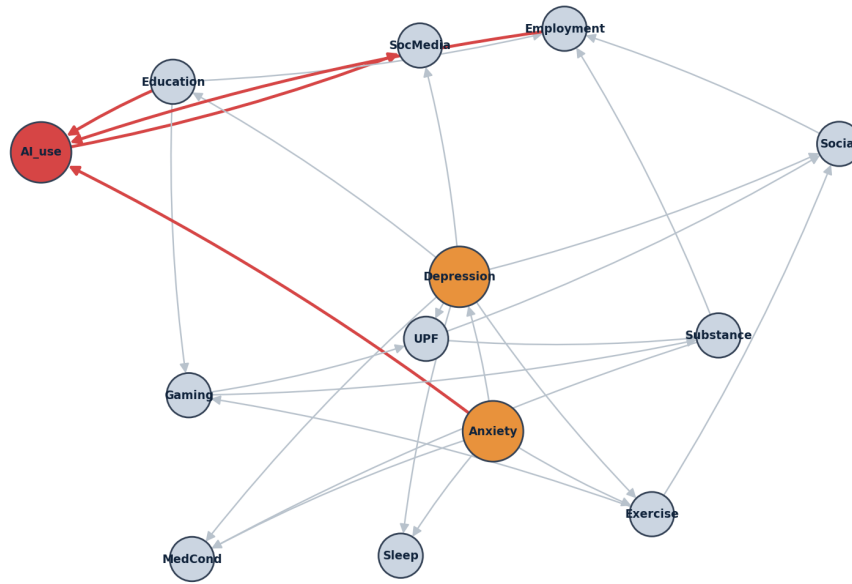


Figure 2. Learned Bayesian network, young adults (n=45,366). Red = AI-use; orange = distress outcomes; red edges trace AI-use's direct links, which reach both Anxiety and Depression.

Age group	% AI users	$\rho(\text{freq, MHQ})$	AI-use's direct neighbours
18–24	93%	–0.091	Anxiety, Depression, Education, Employment, SocMedia
25–34	81%	–0.052	Education, Employment
35–44	71%	–0.051	Education, Employment
45–54	65%	–0.040	Education, Employment
55–64	55%	–0.011	Education, Exercise
65+	42%	–0.008	Education

Figure-of-merit: the AI–distress association fades monotonically with age and only survives as a direct network edge in the youngest group.

4. Is it just screen time? AI vs social-media vs gaming

If heavy AI-use were merely a marker of a heavy-screen lifestyle, its association with distress should disappear once social-media and gaming are included. It does not. The adjusted odds of high anxiety for heavy (vs moderate) AI use barely move — from 1.52 to 1.50 — after adding both other screens and eight confounders.

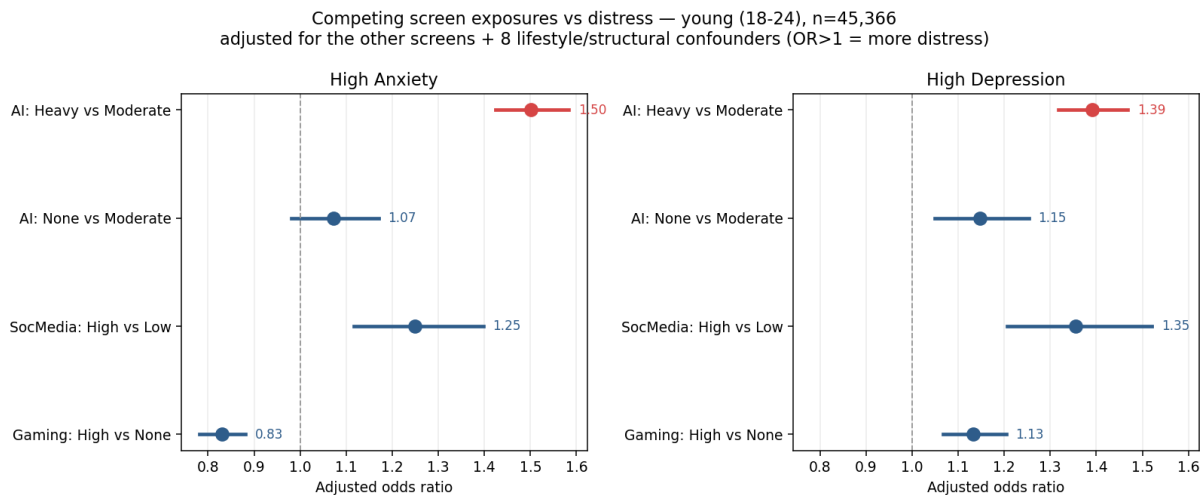


Figure 3. Competing screen exposures vs distress, young adults. Adjusted odds ratios (>1 = more distress). Heavy AI-use (red) is the strongest screen risk for anxiety; gaming is protective; moderate social-media is protective.

In the Bayesian network with all three screens competing, AI-use keeps direct edges to *both* anxiety and depression; social-media connects to depression only; gaming connects to neither distress node (it clusters with exercise, diet and substance use). AI carries a distinct signal.

5. The decisive result: purpose beats frequency

Splitting young AI users by *what they use AI for* reveals two opposite populations. Those who report tool-like benefits are **less** anxious than average; those who use AI for companionship or report dependence / isolation are **much more** anxious. The effect sizes dwarf anything frequency produced (Cohen's d up to +0.55).

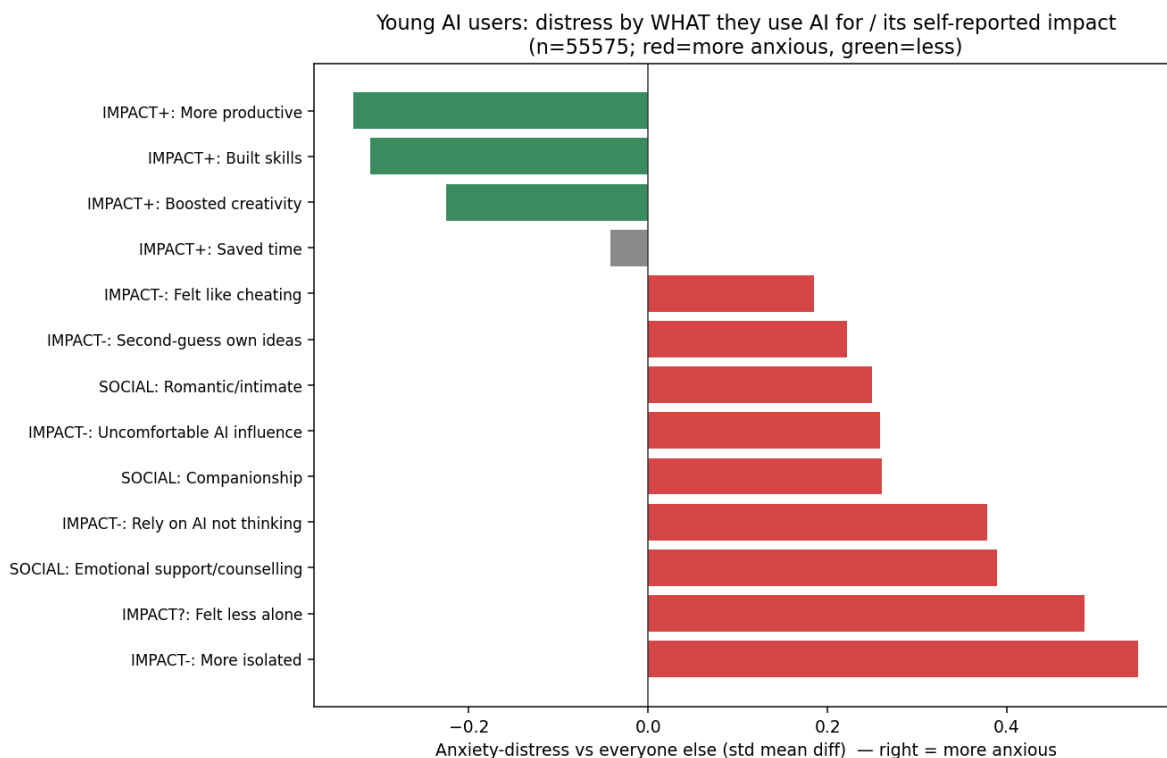


Figure 4. Anxiety by AI-use purpose / self-reported impact, young users (n=55,575). Green = lower anxiety than peers; red = higher. Tool-use protects; companion-use and reported dependence/isolation carry the risk.

Put purpose and frequency in one model and frequency collapses:

Predictor of high anxiety (young users)	Odds ratio
Uses AI for emotional support / companionship / romantic	1.56
“Rely on AI instead of thinking”	1.61
Feels “more isolated” because of AI	1.84
AI-use frequency (per step)	1.04

Frequency’s earlier signal was largely a *proxy* for companion-mode use, which is more common among heavy users.

6. The companion link: independent of support, present at all ages

We tested whether the companion-use penalty concentrates among young people with weak real-world support (feeling loved; family closeness). It does not — companion-use raises anxiety at every support level. Companion-use (OR 1.66) and feeling unloved (OR 1.71) are strong, largely independent risk markers.

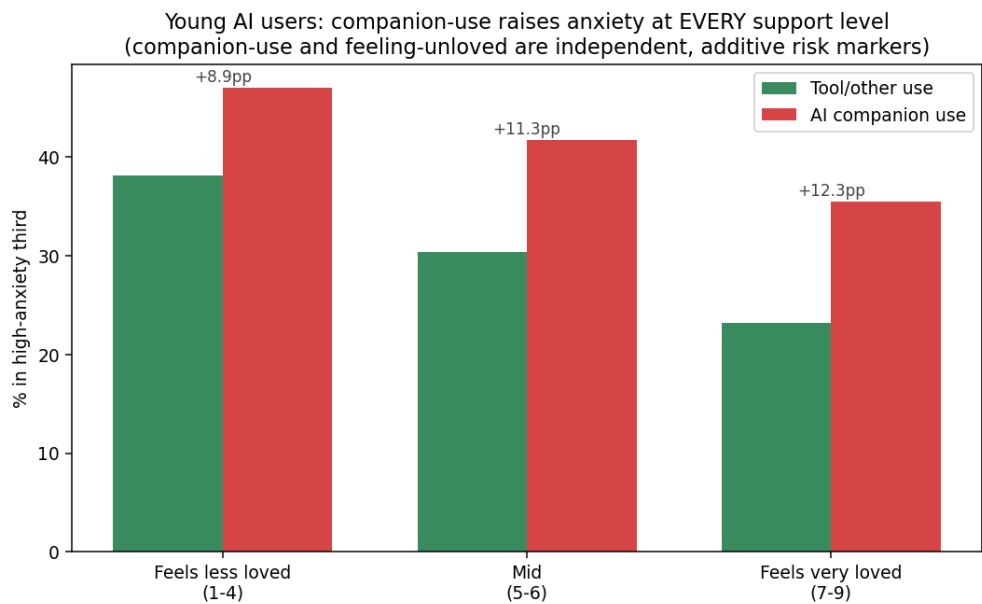


Figure 5. Companion-use raises anxiety at every level of feeling loved. Companion-use and low support are independent, additive risk markers.

Extending across the lifespan resolves the earlier ‘young-specific’ puzzle. The *prevalence* of companion-use falls threefold with age, but the companion–anxiety *association* is present at every age. The young are not uniquely vulnerable to AI — they are simply far more exposed to the companion mode that carries the signal.

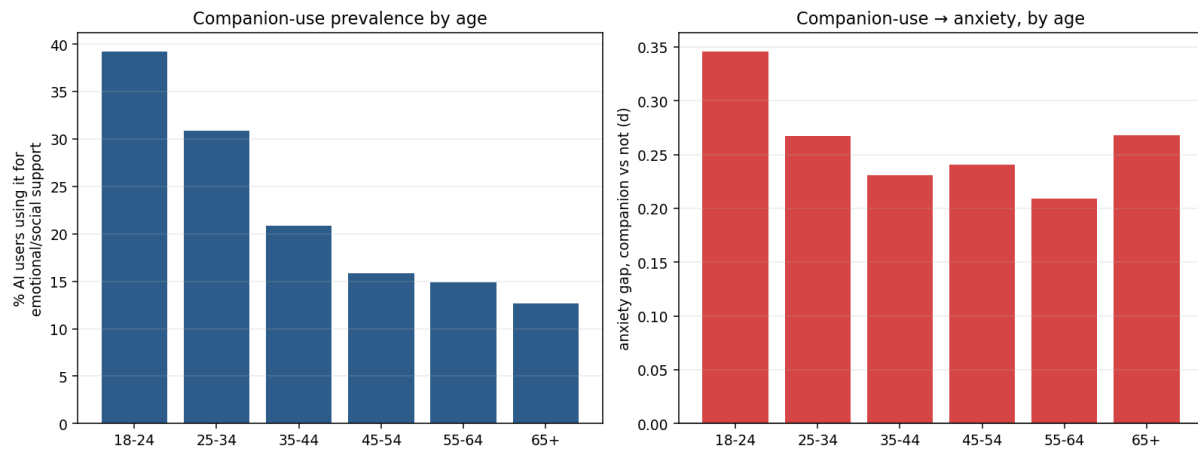


Figure 6. Left: share of AI users who use it for emotional/social support, by age (39% → 13%). Right: the companion→anxiety gap (Cohen's d) is present at all ages.

7. Synthesis and limitations

- AI-use **frequency** has only a weak, young-concentrated link to mental health, shaped as an inverted-U (moderate best, heavy worst).
- That link is mostly a **proxy for purpose**: tool-use is benign-to-protective; companion-use is the real risk marker.
- The companion–distress relationship is **not explained by general screen time**, is **independent of real-world support**, and is **present at all ages**.
- The young carry far more of it because they adopt the companion mode three times as often.

Limitations. All analyses are cross-sectional; causal direction is not identified. The distress proxies are drawn from the same item pool as MHQ (the independent PHQ-9/GAD-7 fields were empty here). Self-reported impact items ('more isolated', 'rely on AI') may be coloured by current mood. The most plausible reading — that anxious or lonely people are drawn to AI companionship — is at least as consistent with the data as the reverse.

Recommended next step. A longitudinal or instrumented design measuring AI-use purpose (not just frequency) over time would be required to establish direction. Of the available data, purpose-of-use should be the primary exposure in any further modelling.